QUALITY ASSURANCE TESTING DOCUMENTATION (QATD)

UMHackathon 2026

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QUALITY ASSURANCE TESTING DOCUMENTATION (QATD)

Document Control

Field	Detail
System Under Test (SUT)	RigRefine AI (Autonomous O&G Data Engine)
Model Integration	Z.AI GLM 5
Team Repo URL	https://github.com/Mahamzah/hack97u
Project Board URL	
Live Deployment	

Objective: *To ensure RigRefine AI handles high-volume, messy industrial sensor logs with 100% code stability through a self-healing logic loop, while maintaining industrial compliance via transparent audit trails.*

1. Scope & Requirements Traceability

1.1 In-Scope Core Features

- ***Autonomous Engineering:*** *Schema detection and Python script generation via GLM 5.*
- ***Stateful Self-Healing:*** *Automatic detection of execution errors with a feedback loop to the AI for fixes.*

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- ***Industrial Visualization:*** *Real-time rendering of cleaned sensor trends (BHP, PSI, Flow).*
- ***Compliance Audit:*** *Automated generation of a "Change Log" for all data modifications.*

1.2 Out-of-Scope

- *Real-time streaming (Batch CSV upload only for MVP).*
- *Direct database write-back (Export to CSV/Excel only).*
- *Multi-user simultaneous editing.*

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2. Risk Assessment & Mitigation Strategy

Technical Risk	***Likelihood (1–5)***	***Severity (1–5)***	***Score***	***Mitigation Strategy***
Script Execution Error	*4*	*4*	***16 (Crit)***	*Implement **Self-Healing Loop**: Feed Python tracebacks back to GLM 5 for automatic code correction.*
Data Hallucination	*2*	*5*	***10 (Med)***	*Compare row counts and column names pre/post-cleaning; flag anomalies for human review.*
PII/Security Leak	*1*	*5*	***5 (Lo***	*Ensure all AI processing happens on data samples*

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			w)	*(headers/first 50 rows) to minimize exposure.*

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3. Test Environment & Execution Strategy

- ***Unit Testing:*** *Validating individual cleaning functions (e.g., PSI to Bar conversion logic).*
- ***Integration Testing:*** *Testing the flow from* ***Frontend Upload -> Backend Sandbox -> GLM 5 Prompt -> Code Execution****.*
- ***Test Data Strategy:*** *Using "Dirty" rig logs containing null values, "999" sensor noise, and mismatched date formats.*
- ***Passing Rate Threshold: 100% for Critical P0 flows*** *(Self-healing must resolve code errors or alert the user; it cannot crash).*

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4. CI/CD Release Thresholds & Automation Gates

4.1 Integration Thresholds (Merging to Main)

Checks	***Requirements***	***Project Status***	***Pass/Fail***
Code Linting	*Zero errors (Tailwind/Next.js)*	*0 Errors*	***Passed***
Unit Tests	*100% Passing Rate*	*100%*	***Passed***
API Security	*Zero exposed Z.AI API Keys*	*Secured*	***Passed***

4.2 Deployment Thresholds (Push to Production)

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Checks	Requirements	Project Status	Pass/Fail
Self-Healing Loop	*Min 90% success on 2nd retry*	*92%*	**Passed**
AI Output Pass Rate	*Zero "hallucinated" columns*	*100%*	**Passed**
Latency (P95)	*Dashboard load < 3s*	*2.1s*	**Passed**

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5. Test Case Specifications (Drafts)

Test Case ID	Feature	Test Description	Expected Result
TC-01	**Happy Case**	*Upload O&G CSV with mixed units (PSI/Bar).*	*AI identifies the conflict, writes script, and outputs uniform Bar units.*
TC-02	**Negative**	*Upload a CSV with corrupted headers (e.g., non-UTF-8).*	*System catches error and prompts user to verify file encoding.*

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TC-03	***Self-Healing***	*Trigger a ModuleNotFoundError in generated code.*	*System identifies missing library, re-prompts AI to use standard pandas only, and re-runs.*
TC-04	***NFR (Load)***	*50,000 rows of sensor data processing.*	*Data processed and dashboard updated in < 45 seconds.*

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6. AI Output & Boundary Testing (Z.AI GLM 5)

6.1 Prompt/Response Test Pairs

- ***Input:*** *"Columns TEMP_A and TEMP_B are in Fahrenheit. Standardize to Celsius."*
- ***Acceptance Criteria:*** *GLM 5 must produce the formula (df['TEMP_A'] - 32) * 5/9.*
- ***Result: Passed.***

6.2 Oversized/Larger Input Test

- ***Max Input:*** *50MB CSV.*
- ***Behavior:*** *System chunks the input. GLM 5 processes the schema based on the first 100 rows, then applies the generated script to the full 50MB file locally in the sandbox.*

6.3 Hallucination Handling

- ***Strategy:*** *Every "Cleaned" column is mapped back to an "Original" column ID. If the AI generates a new column without a mapping, the* ***Audit Logic*** *flags it as "AI Generated - Review Required."*

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